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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 24th June Evening Shift

JEE Main Question Paper with Answer Key

24 Jun 2022	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	21 questions
Mathematics	24 questions
Physics	25 questions
Out of syllabus	20 questions
Answer key	at the end

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Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

120 g of an organic compound that contains only carbon and hydrogen gives 330 g of CO_2 and 270 g of water on complete combustion. The percentage of carbon and hydrogen, respectively are

(A) 25 and 75 (B) 40 and 60 (C) 60 and 40 (D) 75 and 25

Open solution

Q2. MCQ (Single Correct Answer)

The energy of one mole of photons of radiation of wavelength 300 nm is

(Given : h = 6.63×10^{-34} J s, $N_A = 6.02 \times 10^{23}$ mol^{-1}, c = 3×10^8 m s^{-1})

(A) 235 kJ mol^{-1} (B) 325 kJ mol^{-1} (C) 399 kJ mol^{-1} (D) 435 kJ mol^{-1}

Open solution

Q3. MCQ (Single Correct Answer)

The correct order of bond orders of C_2^{2-}, N_2^{2-} and O_2^{2-} is, respectively

(A) $C_2^{2-} < N_2^{2-} < O_2^{2-}$ (B) $O_2^{2-} < N_2^{2-} < C_2^{2-}$

(C) $C_2^{2-} < O_2^{2-} < N_2^{2-}$ (D) $N_2^{2-} < C_2^{2-} < O_2^{2-}$

Open solution

Q4. MCQ (Single Correct Answer)

At 25°C and 1 atm pressure, the enthalpies of combustion are as given below :

Substance	H_2	C (graphite)	$C_2H_6(g)$
$\frac{\Delta_c H^\Theta}{kJ\ mol^{-1}}$	-286.0	-394.0	-1560.0

The enthalpy of formation of ethane is

(A) +54.0 kJ mol^{-1} (B) -68.0 kJ mol^{-1} (C) -86.0 kJ mol^{-1} (D) +97.0 kJ mol^{-1}

Open solution

Q5. MCQ (Single Correct Answer)

For a first order reaction, the time required for completion of 90% reaction is 'x' times the half life of the reaction. The value of 'x' is

(Given : ln 10 = 2.303 and log 2 = 0.3010)

(A) 1.12 (B) 2.43 (C) 3.32 (D) 33.31

Open solution

Q6. MCQ (Single Correct Answer)

Metals generally melt at very high temperature. Amongst the following, the metal with the highest melting point will be :

(A) Hg (B) Ag (C) Ga (D) Cs

Open solution

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Q7. MCQ (Single Correct Answer)

PCl_5 is well known, but NCl_5 is not. because,

(A) nitrogen is less reactive than phosphorus.

(B) nitrogen doesn't have d-orbitals in its valence shell.

(C) catenation tendency is weaker in nitrogen than phosphorus.

(D) size of phosphorus is larger than nitrogen.

Open solution

Q8. MCQ (Single Correct Answer)

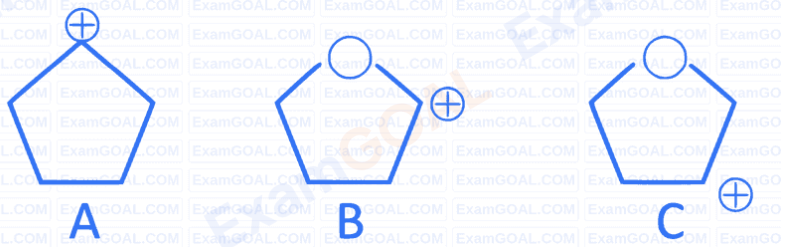
Transition metal complex with highest value of crystal field splitting (Δ_0) will be :

(A) $[Cr(H_2O)_6]^{3+}$ (B) $[Mo(H_2O)_6]^{3+}$ (C) $[Fe(H_2O)_6]^{3+}$ (D) $[Os(H_2O)_6]^{3+}$

Open solution

Q9. MCQ (Single Correct Answer)

Arrange the following carbocations in decreasing order of stability.



(A) B > A > C (B) A > B > C (C) C > B > A (D) C > A > B

Open solution

Q10. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The presence of weaker π-bonds makes alkenes less stable than alkanes.

Statement II : The strength of the double bond is greater than that of carbon-carbon single bond.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Open solution

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Q11. MCQ (Single Correct Answer)

Open solution

Which of the following reagents / reactions will convert 'A' to 'B' ?



(A) PCC oxidation

(B) Ozonolysis

(C) BH_3, $H_2O_2/^{-}OH$ followed by PCC oxidation

(D) HBr, hydrolysis followed by oxidation by $K_2Cr_2O_7$.

Q12. MCQ (Single Correct Answer)

Open solution

Hex-4-ene-2-ol on treatment with PCC gives 'A'. 'A' on reaction with sodium hypoiodite gives 'B', which on further heating with soda lime gives 'C'. The compound 'C' is :

(A) 2-pentene

(B) proponaldehyde

(C) 2-butene

(D) 4-methylpent-2-ene

Q13. MCQ (Single Correct Answer)

Open solution

The conversion of propan-1-ol to n-butylamine involves the sequential addition of reagents. The correct sequential order of reagents is

(A) (i) $SOCl_2$ (ii) KCN (iii) H_2/Ni, Na(Hg)/C_2H_5OH

(B) (i) HCl (ii) H_2/Ni, Na(Hg)/C_2H_5OH

(C) (i) $SOCl_2$ (ii) KCN (iii) CH_3NH_2

(D) (i) HCl (ii) CH_3NH_2

Q14. MCQ (Single Correct Answer)

Open solution

In the flame test of a mixture of salts, a green flame with blue centre was observed. Which one of the following cations may be present?

(A) Cu^{2+}

(B) Sr^{2+}

(C) Ba^{2+}

(D) Ca^{2+}

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Q15. Numerical

A company dissolves 'x' amount of CO_2 at 298 K in 1 litre of water to prepare soda water. X = _____ $\times 10^{-3}$ g. (nearest integer)

(Given : partial pressure of CO_2 at 298 K = 0.835 bar.

Henry's law constant for CO_2 at 298 K = 1.67 kbar.

Atomic mass of H, C and O is 1, 12, and 16 g mol^{-1}, respectively)

Open solution

Q16. Numerical

PCl_5 dissociates as

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2 and is maintained at 600 K. The equilibrium pressure is 2.46 atm. The equilibrium constant K_p for the dissociation of PCl_5 is _____ $\times 10^{-3}$. (nearest integer)

(Given : R = 0.082 L atm K^{-1} mol^{-1}; Assume ideal gas behaviour)

Open solution

Q17. Numerical

The resistance of a conductivity cell containing 0.01 M KCl solution at 298 K is 1750 Ω. If the conductivity of 0.01 M KCl solution at 298 K is 0.152 $\times 10^{-3}$ S cm^{-1}, then the cell constant of the conductivity cell is _____ $\times 10^{-3}$ cm^{-1}.

Open solution

Q18. Numerical

Manganese (VI) has ability to disproportionate in acidic solution. The difference in oxidation states of two ions it forms in acidic solution is _____.

Open solution

Q19. Numerical

0.2 g of an organic compound was subjected to estimation of nitrogen by Dumas method in which volume of N_2 evolved (at STP) was found to be 22.400 mL. The percentage of nitrogen in the compound is _____. [nearest integer]

(Given : Molar mass of N_2 is 28 g mol^{-1}. Molar volume of N_2 at STP : 22.4 L)

Open solution

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Q20. Numerical



Open solution

Consider the above reaction. The number of π electrons present in the product 'P' is _____.

Q21. Numerical

In alanylglycyl leucyl alanyl valine, the number of peptide linkages is _____.

Open solution

Mathematics 24 questions

Q22. MCQ (Single Correct Answer)

Let $x * y = x^2 + y^3$ and $(x * 1) * 1 = x * (1 * 1)$.

Then a value of $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$ is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

Open solution

Q23. MCQ (Single Correct Answer)

The sum of all the real roots of the equation

$(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is

(A) $\log_e 3$ (B) $-\log_e 3$ (C) $\log_e 6$ (D) $-\log_e 6$

Open solution

Q24. MCQ (Single Correct Answer)

Let the system of linear equations

x + y + αz = 2

3x + y + z = 4

x + 2z = 1

have a unique solution (x*, y*, z*). If (α, x*), (y*, α) and (x*, $-$y*) are collinear points, then the sum of absolute values of all possible values of α is

(A) 4 (B) 3 (C) 2 (D) 1

Open solution

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Q25. MCQ (Single Correct Answer)

Let x, y > 0. If $x^3y^2 = 2^{15}$, then the least value of 3x + 2y is

(A) 30 (B) 32 (C) 36 (D) 40

Open solution

Q26. MCQ (Single Correct Answer)

Let $f(x) = \begin{cases} \frac{\sin(x-[x])}{x-[x]} & , x \in (-2,-1) \\ \max\{2x, 3[|x|]\} & , |x| < 1 \\ 1 & , otherwise \end{cases}$

where [t] denotes greatest integer $\leq$ t. If m is the number of points where f is not continuous and n is the number of points where f is not differentiable, then the ordered pair (m, n) is :

(A) (3, 3) (B) (2, 4) (C) (2, 3) (D) (3, 4)

Open solution

Q27. MCQ (Single Correct Answer)

The value of the integral

$\int_{-\pi/2}^{\pi/2} \frac{dx}{(1+e^x)(\sin^6 x+\cos^6 x)}$ is equal to

(A) 2π (B) 0 (C) π (D) $\frac{\pi}{2}$

Open solution

Q28. MCQ (Single Correct Answer)

A particle is moving in the xy-plane along a curve C passing through the point (3, 3). The tangent to the curve C at the point P meets the x-axis at Q. If the y-axis bisects the segment PQ, then C is a parabola with :

(A) length of latus rectum 3 (B) length of latus rectum 6

(C) focus $\left(\frac{4}{3}, 0\right)$ (D) focus $\left(0, \frac{3}{4}\right)$

Open solution

Q29. MCQ (Single Correct Answer)

Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$, $a > 2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be $6\sqrt{3}$. Then the eccentricity of the ellipse is :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{4}$

Open solution

Q30. MCQ (Single Correct Answer)

Let the area of the triangle with vertices A(1, α), B(α, 0) and C(0, α) be 4 sq. units. If the points (α, $-\alpha$), ($-\alpha$, α) and (α^2, β) are collinear, then β is equal to :

(A) 64 (B) -8 (C) -64 (D) 512

Open solution

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Q31. MCQ (Single Correct Answer)

The number of distinct real roots of the equation

$x^7 - 7x - 2 = 0$ is

(A) 5 (B) 7 (C) 1 (D) 3

Open solution

Q32. MCQ (Single Correct Answer)

A random variable X has the following probability distribution :

X	0	1	2	3	4
P(X)	k	2k	4k	6k	8k

The value of $P(1 < X < 4 \mid X \leq 2)$ is equal to :

(A) $\frac{4}{7}$ (B) $\frac{2}{3}$ (C) $\frac{3}{7}$ (D) $\frac{4}{5}$

Open solution

Q33. MCQ (Single Correct Answer)

If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible value of λ is :

(A) 16 (B) 6 (C) 12 (D) 15

Open solution

Q34. MCQ (Single Correct Answer)

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements :

(S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$

(S2) : The projection of $\hat{a}$ on $(\hat{a} + \hat{b})$ is $\frac{1}{2}$

(A) Only (S1) is true. (B) Only (S2) is true.

(C) Both (S1) and (S2) are true. (D) Both (S1) and (S2) are false.

Open solution

Q35. MCQ (Single Correct Answer)

If $y = \tan^{-1}\left(\sec x^3 - \tan x^3\right), \frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, then

(A) $xy'' + 2y' = 0$ (B) $x^2y'' - 6y + \frac{3\pi}{2} = 0$

(C) $x^2y'' - 6y + 3\pi = 0$ (D) $xy'' - 4y' = 0$

Open solution

Q36. MCQ (Single Correct Answer)

Let λ^* be the largest value of λ for which the function $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ is increasing for all x $\in$ R. Then $f_{\lambda^*}(1) + f_{\lambda^*}(-1)$ is equal to :

(A) 36 (B) 48 (C) 64 (D) 72

Open solution

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Q37. Numerical

Let S = {z ∈ C : |z − 3| ≤ 1 and z(4 + 3i) + $\overline{z}$(4 − 3i) ≤ 24}. If α + iβ is the point in S which is closest to 4i, then 25(α + β) is equal to _____.

Open solution

Q38. Numerical

Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; a, b \in \{1, 2, 3, \ldots . 100\} \right\}$ and let $T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of elements in $\bigcap_{n=1}^{100} T_n$ is _____.

Open solution

Q39. Numerical

The number of 7-digit numbers which are multiples of 11 and are formed using all the digits 1, 2, 3, 4, 5, 7 and 9 is _____.

Open solution

Q40. Numerical

The sum of all the elements of the set $\{\alpha \in \{1, 2, \ldots . . , 100\} : HCF(\alpha, 24) = 1\}$ is _____.

Open solution

Q41. Numerical

The remainder on dividing 1 + 3 + 3^2 + 3^3 + ... + 3^{2021} by 50 is _____.

Open solution

Q42. Numerical

The area (in sq. units) of the region enclosed between the parabola y^2 = 2x and the line x + y = 4 is _____.

Open solution

Q43. Numerical

Let a circle C : $(x - h)^2 + (y - k)^2 = r^2$, k > 0, touch the x-axis at (1, 0). If the line x + y = 0 intersects the circle C at P and Q such that the length of the chord PQ is 2, then the value of h + k + r is equal to _____.

Open solution

Q44. Numerical

Let the hyperbola $H : \frac{x^2}{a^2} - y^2 = 1$ and the ellipse $E : 3x^2 + 4y^2 = 12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E. If e_H and e_E are the eccentricities of H and E respectively, then the value of $12\left(e_H^2 + e_E^2\right)$ is equal to _____.

Open solution

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Q45. Numerical

Let P_1 be a parabola with vertex (3, 2) and focus (4, 4) and P_2 be its mirror image with respect to the line x + 2y = 6. Then the directrix of P_2 is x + 2y = _____.

Open solution

Physics 25 questions

Q46. MCQ (Single Correct Answer)

Identify the pair of physical quantities that have same dimensions:

(A) velocity gradient and decay constant
(B) Wien's constant and Stefan constant
(C) angular frequency and angular momentum
(D) wave number and Avogadro number

Open solution

Q47. MCQ (Single Correct Answer)

The distance between Sun and Earth is R. The duration of year if the distance between Sun and Earth becomes 3R will be :

(A) $\sqrt{3}$ years
(B) 3 years
(C) 9 years
(D) $3\sqrt{3}$ years

Open solution

Q48. MCQ (Single Correct Answer)

A stone of mass m, tied to a string is being whirled in a vertical circle with a uniform speed. The tension in the string is

(A) the same throughout the motion.
(B) minimum at the highest position of the circular path.
(C) minimum at the lowest position of the circular path.
(D) minimum when the rope is in the horizontal position.

Open solution

Q49. MCQ (Single Correct Answer)

Two identical charged particles each having a mass 10 g and charge 2.0×10^{-7}C are placed on a horizontal table with a separation of L between them such that they stay in limiting equilibrium. If the coefficient of friction between each particle and the table is 0.25, find the value of L. [Use g = 10 ms^{-2}]

(A) 12 cm
(B) 10 cm
(C) 8 cm
(D) 5 cm

Open solution

Q50. MCQ (Single Correct Answer)

Two massless springs with spring constants 2 k and 9 k, carry 50 g and 100 g masses at their free ends. These two masses oscillate vertically such that their maximum velocities are equal. Then, the ratio of their respective amplitudes will be :

(A) 1 : 2
(B) 3 : 2
(C) 3 : 1
(D) 2 : 3

Open solution

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Q51. MCQ (Single Correct Answer)

What will be the most suitable combination of three resistors A = 2Ω, B = 4Ω, C = 6Ω so that $\left(\frac{22}{3}\right)\Omega$ is equivalent resistance of combination?

Open solution

(A) Parallel combination of A and C connected in series with B.

(B) Parallel combination of A and B connected in series with C.

(C) Series combination of A and C connected in parallel with B.

(D) Series combination of B and C connected in parallel with A.

Q52. MCQ (Single Correct Answer)

The soft-iron is a suitable material for making an electromagnet. This is because soft-iron has

Open solution

(A) low coercivity and high retentivity.

(B) low coercivity and low permeability.

(C) high permeability and low retentivity.

(D) high permeability and high retentivity.

Q53. MCQ (Single Correct Answer)

A proton, a deuteron and an α-particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is :

Open solution

(A) $1:\sqrt{2}:\sqrt{2}$ (B) $1:1:\sqrt{2}$ (C) $\sqrt{2}:1:1$ (D) $1:\sqrt{2}:1$

Q54. MCQ (Single Correct Answer)

Given below are two statements :

Open solution

Statement I : The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.

Statement II : In ac circuit, the average power delivered by the source never becomes zero.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q55. MCQ (Single Correct Answer)

Potential energy as a function of r is given by $U = \frac{A}{r^{10}} - \frac{B}{r^5}$, where r is the interatomic distance, A and B are positive constants. The equilibrium distance between the two atoms will be :

Open solution

(A) $\left(\frac{A}{B}\right)^{\frac{1}{5}}$ (B) $\left(\frac{B}{A}\right)^{\frac{1}{5}}$

(C) $\left(\frac{2A}{B}\right)^{\frac{1}{5}}$ (D) $\left(\frac{B}{2A}\right)^{\frac{1}{5}}$

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Q56. MCQ (Single Correct Answer)

An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to : [Use g = 10 ms^{-2}].

(A) 1 : 1 (B) $\sqrt{2} : \sqrt{3}$ (C) $\sqrt{3} : \sqrt{2}$ (D) 2 : 3

Open solution

Q57. MCQ (Single Correct Answer)

A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be :

(A) 7.5 rad (B) 15 rad (C) 20 rad (D) 30 rad

Open solution

Q58. MCQ (Single Correct Answer)

A 100 g of iron nail is hit by a 1.5 kg hammer striking at a velocity of 60 ms^{-1}. What will be the rise in the temperature of the nail if one fourth of energy of the hammer goes into heating the nail?

[Specific heat capacity of iron = 0.42 $Jg^{-1}\,{}^{\circ}C^{-1}$]

(A) 675°C (B) 1600°C (C) 16.07°C (D) 6.75°C

Open solution

Q59. MCQ (Single Correct Answer)

If the charge on a capacitor is increased by 2 C, the energy stored in it increases by 44%. The original charge on the capacitor is (in C)

(A) 10 (B) 20 (C) 30 (D) 40

Open solution

Q60. MCQ (Single Correct Answer)

A long cylindrical volume contains a uniformly distributed charge of density ρ. The radius of cylindrical volume is R. A charge particle (q) revolves around the cylinder in a circular path. The kinetic energy of the particle is :

(A) $\frac{\rho q R^2}{4\varepsilon_0}$ (B) $\frac{\rho q R^2}{2\varepsilon_0}$

(C) $\frac{q\rho}{4\varepsilon_0 R^2}$ (D) $\frac{4\varepsilon_0 R^2}{q\rho}$

Open solution

Q61. MCQ (Single Correct Answer)

An electric bulb is rated as 200 W. What will be the peak magnetic field at 4 m distance produced by the radiations coming from this bulb? Consider this bulb as a point source with 3.5% efficiency.

(A) 1.19×10^{-8}T (B) 1.71×10^{-8}T (C) 0.84×10^{-8}T (D) 3.36×10^{-8}T

Open solution

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Q62. MCQ (Single Correct Answer)

The light of two different frequencies whose photons have energies 3.8 eV and 1.4 eV respectively, illuminate a metallic surface whose work function is 0.6 eV successively. The ratio of maximum speeds of emitted electrons for the two frequencies respectively will be :

(A) 1 : 1 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

Open solution

Q63. MCQ (Single Correct Answer)

Two light beams of intensities in the ratio of 9 : 4 are allowed to interfere. The ratio of the intensity of maxima and minima will be :

(A) 2 : 3 (B) 16 : 81 (C) 25 : 169 (D) 25 : 1

Open solution

Q64. MCQ (Single Correct Answer)

In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level :

(A) All K, P and E increase.
(B) K decreases, P and E increase.
(C) P decreases, K and E increase.
(D) K increases, P and E decrease.

Open solution

Q65. Numerical

A body is projected from the ground at an angle of 45° with the horizontal. Its velocity after 2s is 20 ms^{-1}. The maximum height reached by the body during its motion is _____ m. (use g = 10 ms^{-2})

Open solution

Q66. Numerical

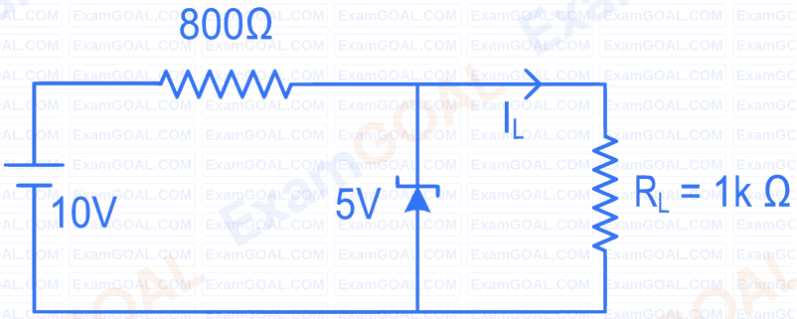
Two travelling waves of equal amplitudes and equal frequencies move in opposite directions along a string. They interfere to produce a stationary wave whose equation is given by $y = (10 \cos \pi x \sin \frac{2\pi t}{T})$ cm

The amplitude of the particle at $x = \frac{4}{3}$ cm will be _____ cm.

Open solution

Q67. Numerical

In the given circuit, the value of current I_L will be _____ mA. (When R_L = 1kΩ)



Open solution

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Q68. Numerical

A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ cm. The thickness of the glass slab is _____ cm.

Open solution

Q69. Numerical

A circular coil of 1000 turns each with area $1m^2$ is rotated about its vertical diameter at the rate of one revolution per second in a uniform horizontal magnetic field of 0.07T. The maximum voltage generation will be _____ V.

Open solution

Q70. Numerical

A monoatomic gas performs a work of $\frac{Q}{4}$ where Q is the heat supplied to it. The molar heat capacity of the gas will be _____ R during this transformation. Where R is the gas constant.

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Which of the following chemical reactions represents Hall-Heroult Process?

(A) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$

(B) $2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$

(C) $FeO + CO \rightarrow Fe + CO_2$

(D) $2[Au(CN)_2]^-_{(aq)} + Zn(s) \rightarrow 2Au(s) + [Zn(CN_4)]^{2-}$

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

In the industrial production of which of the following, molecular hydrogen is obtained as a byproduct?

(A) NaOH (B) NaCl (C) Na metal (D) Na_2CO_3

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following compounds is used as a chemical in certain type of fire extinguishers?

(A) Baking soda (B) Soda ash (C) Washing soda (D) Caustic soda

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Some gases are responsible for heating of atmosphere (green house effect). Identify from the following the gaseous species which does not cause it.

(A) CH_4 (B) O_3 (C) H_2O (D) N_2

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

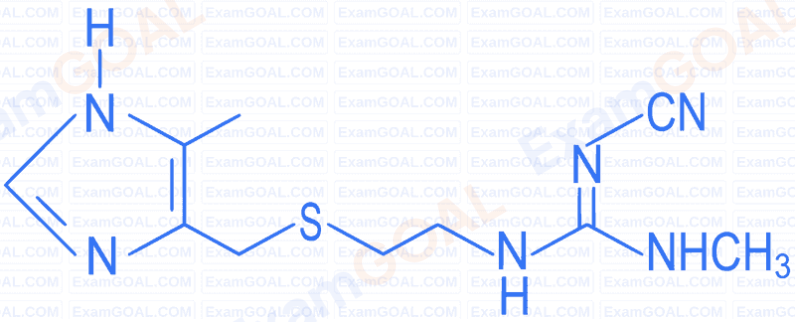
Which of the following is not an example of a condensation polymer?

(A) Nylon 6,6 (B) Dacron (C) Buna-N (D) Silicone

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

The structure shown below is of which well-known drug molecule?



(A) Ranitidine (B) Seldane (C) Cimetidine (D) Codeine

Open solution

Q77. Numerical · **Chemistry**

At 300 K, a sample of 3.0 g of gas A occupies the same volume as 0.2 g of hydrogen at 200 K at the same pressure. The molar mass of gas A is _____ g mol^{-1}. (nearest integer) Assume that the behaviour of gases as ideal.

(Given : The molar mass of hydrogen (H_2) gas is 2.0 g mol^{-1}.)

Open solution

Q78. Numerical · **Chemistry**

When 200 mL of 0.2 M acetic acid is shaken with 0.6 g of wood charcoal, the final concentration of acetic acid after adsorption is 0.1 M. The mass of acetic acid adsorbed per gram of carbon is _____ g.

Open solution

Q79. Numerical · **Chemistry**

(a) Baryte, (b) Galena, (c) Zinc blende and (d) Copper pyrites. How many of these minerals are sulphide based?

Open solution

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Q80. MCQ (Single Correct Answer) · **Mathematics**

$\lim\limits_{n\to\infty}\left(\frac{n^2}{(n^2+1)(n+1)}+\frac{n^2}{(n^2+4)(n+2)}+\frac{n^2}{(n^2+9)(n+3)}+\ldots.+\frac{n^2}{(n^2+n^2)(n+n)}\right)$ is equal to :

(A) $\frac{\pi}{8}+\frac{1}{4}\log_e 2$

(B) $\frac{\pi}{4}+\frac{1}{8}\log_e 2$

(C) $\frac{\pi}{4}-\frac{1}{8}\log_e 2$

(D) $\frac{\pi}{8}+\log_e\sqrt{2}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

The number of solutions of the equation

$\cos\left(x+\frac{\pi}{3}\right)\cos\left(\frac{\pi}{3}-x\right)=\frac{1}{4}\cos^2 2x, x\in[-3\pi,3\pi]$ is :

(A) 8

(B) 5

(C) 6

(D) 7

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let the points on the plane P be equidistant from the points $(-4, 2, 1)$ and $(2, -2, 3)$. Then the acute angle between the plane P and the plane $2x + y + 3z = 1$ is :

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{5\pi}{12}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Consider the following statements:

A : Rishi is a judge.

B : Rishi is honest.

C : Rishi is not arrogant.

The negation of the statement "if Rishi is a judge and he is not arrogant, then he is honest" is

(A) $B\rightarrow(A\vee C)$

(B) $(\sim B)\wedge(A\wedge C)$

(C) $B\rightarrow((\sim A)\vee(\sim C))$

(D) $B\rightarrow(A\wedge C)$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

The slope of normal at any point (x, y), $x > 0, y > 0$ on the curve $y = y(x)$ is given by $\frac{x^2}{xy-x^2y^2-1}$. If the curve passes through the point (1, 1), then $e \cdot y(e)$ is equal to

(A) $\frac{1-\tan(1)}{1+\tan(1)}$

(B) $\tan(1)$

(C) 1

(D) $\frac{1+\tan(1)}{1-\tan(1)}$

Open solution

Q85. Numerical · **Mathematics**

In an examination, there are 10 true-false type questions. Out of 10, a student can guess the answer of 4 questions correctly with probability $\frac{3}{4}$ and the remaining 6 questions correctly with probability $\frac{1}{4}$. If the probability that the student guesses the answers of exactly 8 questions correctly out of 10 is $\frac{27k}{4^{10}}$, then k is equal to _____.

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Q86. MCQ (Single Correct Answer) · **Physics**

A Carnot engine takes 5000 kcal of heat from a reservoir at 727°C and gives heat to a sink at 127°C. The work done by the engine is

(A) 3×10^6 J (B) Zero (C) 12.6×10^6 J (D) 8.4×10^6 J

Open solution

Q87. Numerical · **Physics**

An antenna is placed in a dielectric medium of dielectric constant 6.25. If the maximum size of that antenna is 5.0 mm, it can radiate a signal of minimum frequency of _____ GHz.

(Given μ_r = 1 for dielectric medium)

Open solution

Q88. Numerical · **Physics**

A potentiometer wire of length 10 m and resistance 20 Ω is connected in series with a 25 V battery and an external resistance 30 Ω. A cell of emf E in secondary circuit is balanced by 250 cm long potentiometer wire. The value of E (in volt) is $\frac{x}{10}$. The value of x is _____.

Open solution

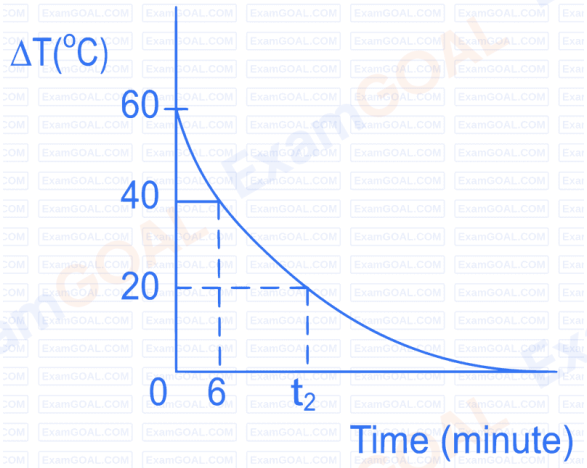
Q89. Numerical · **Physics**

A sample contains 10^{-2} kg each of two substances A and B with half lives 4 s and 8 s respectively. The ratio of their atomic weights is 1 : 2. The ratio of the amounts of A and B after 16 s is $\frac{x}{100}$. The value of x is _____.

Open solution

Q90. Numerical · **Physics**

In an experiment to verify Newton's law of cooling, a graph is plotted between the temperature difference (ΔT) of the water and surroundings and time as shown in figure. The initial temperature of water is taken as 80°C. The value of t_2 as mentioned in the graph will be _____.



Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **B**	4. **C**	5. **C**
6. **B**	7. **B**	8. **D**	9. **A**	10. **A**
11. **C**	12. **C**	13. **A**	14. **A**	15. **1221**
16. **1107**	17. **266**	18. **3**	19. **14**	20. **2**
21. **4**				

Mathematics

22. **B**	23. **B**	24. **C**	25. **D**	26. **C**
27. **C**	28. **A**	29. **A**	30. **C**	31. **D**
32. **A**	33. **A**	34. **C**	35. **B**	36. **D**
37. **80**	38. **100**	39. **576**	40. **1633**	41. **4**
42. **18**	43. **7**	44. **42**	45. **10**	

Physics

46. **A**	47. **D**	48. **B**	49. **A**	50. **B**
51. **B**	52. **C**	53. **D**	54. **C**	55. **C**
56. **B**	57. **B**	58. **C**	59. **A**	60. **A**
61. **B**	62. **B**	63. **D**	64. **B**	65. **20**
66. **5**	67. **5**	68. **12**	69. **440**	70. **2**

Out of syllabus

71. **B**	72. **A**	73. **A**	74. **D**	75. **C**
76. **C**	77. **45**	78. **2**	79. **3**	80. **A**
81. **D**	82. **C**	83. **B**	84. **D**	85. **479**
86. **C**	87. **6**	88. **25**	89. **25**	90. **16**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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